

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – III

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Part A - Program

- a) **Aim:** Create methods `add()`, `multiply()`, `subtract()`, `divide()` with suitable parameters and call these methods using concept of C# delegate.

Algorithm:

1. Start the program.
2. Create a delegate and define calculator methods (Add, Sub, Mul, Div).
3. Create an object of the Calculator class.
4. Assign each method to the delegate and invoke it.
5. Display the results and stop the program.

Code:

```
using System;
delegate void Operation(int a, int b);
class Calculator
{
    public void Add(int a, int b)
    {
        Console.WriteLine("Addition = " + (a + b));
    }
    public void Sub(int a, int b)
    {
        Console.WriteLine("Subtraction = " + (a - b));
    }
    public void Multi(int a, int b)
    {
        Console.WriteLine("Multiplication = " + (a * b));
    }
    public void Div(int a, int b)
    {
        Console.WriteLine("Division = " + (a / b));
    }
}
class Program
{
    static void Main()
    {
        Calculator c = new Calculator();
        Operation obj;
```

```

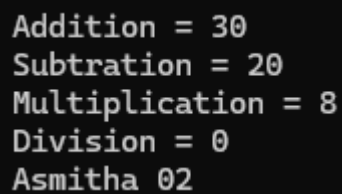
        obj = c.Add;
        obj(10, 20);

        obj = c.Sub;
        obj(40, 20);

        obj = c.Multi;
        obj(4, 2);

        obj = c.Div;
        obj(10, 20);
        Console.WriteLine("Asmitha 02");
    }
}

```

Output:


```

Addition = 30
Subtration = 20
Multiplication = 8
Division = 0
Asmitha 02

```

b) Aim: Write a program using multicast delegate.**Algorithm:**

1. Start the program.
2. Create a delegate and define the methods Welcome(), Learn(), and Thanks().
3. Create an object of the Demo class.
4. Add all methods to the delegate and invoke it.
5. Display the messages and stop the program.

Code:

```

using System;
class BankAccount
{
    public int AccountNumber { get; set; }
    private double balance;
    public double Balance
    {
        get
        {
            return balance;
        }
        set
        {
            if (balance >= 0)
                balance = value;
            else
                Console.WriteLine("Balance cannot be negative.");
        }
    }
}

```

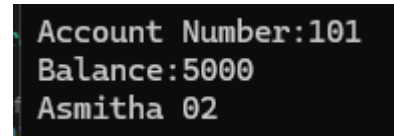
```

    }
}
class Program
{
    static void Main()
    {
        BankAccount b = new BankAccount();
        b.AccountNumber = 101;
        b.Balance = 5000;

        Console.WriteLine("Account Number:" + b.AccountNumber);
        Console.WriteLine("Balance:" + b.Balance);

        b.Balance = -1000;
    }
}

```

Output:


```

Account Number:101
Balance:5000
Asmitha 02

```

c) Aim: Create a class BankAccount with AccountNumber and Balance. Implement property for Balance with validation (Balance cannot be negative).

Algorithm:

1. Start the program.
2. Create the BankAccount class with AccountNumber and Balance properties.
3. Create an object of the BankAccount class.
4. Assign and display the account number and balance.
5. Update the balance, display the details, and stop the program.

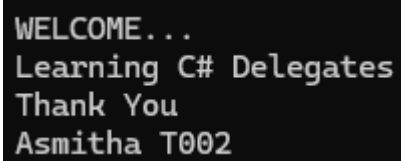
Code:

```

using System;
delegate void Message();
class Demo
{
    public void Welcome()
    {
        Console.WriteLine("WELCOME...");
    }
    public void Learn()
    {
        Console.WriteLine("Learning C# Delegates");
    }
    public void Thanks()
    {
        Console.WriteLine("Thank You");
    }
}

```

```
    }  
}  
public class Program  
{  
    public static void Main(string[] args)  
    {  
        Demo d = new Demo();  
        Message m;  
  
        m = d.Welcome;  
        m += d.Learn;  
        m += d.Thanks;  
        m();  
  
        Console.Write("Asmitha T002");  
    }  
}
```

Output:A screenshot of a terminal window with a black background and white text. The output consists of four lines: 'WELCOME...', 'Learning C# Delegates', 'Thank You', and 'Asmitha T002'.

```
WELCOME...  
Learning C# Delegates  
Thank You  
Asmitha T002
```